

From Enhancement to Transformation: Instructional Design Implications of AI Tool Affordances Through a SAMR Lens in K-12 Education

Abstract

Artificial intelligence (AI) tools are rapidly expanding in K-12 education, yet systematic analyses linking their structural design features to pedagogical frameworks remain limited. This study examines the pedagogical potential of 26 widely used AI educational tools by integrating affordance theory with the Substitution, Augmentation, Modification, Redefinition (SAMR) model. Rather than assessing enacted classroom implementation, the analysis focuses on structural affordances and their alignment with degrees of task change. Using directed content analysis and crisp-set Qualitative Comparative Analysis (csQCA), core affordance indicators were coded and examined configurationally. Results indicate that 84% of the tools demonstrated affordance configurations consistent with Modification- or Redefinition-level task change, 15% aligned with Augmentation-level classifications, while none were classified as purely Substitution-level based on their highest configuration. Reduced-model QCA identified recurring combinations of generative, dialogic, and adaptive affordances associated with higher degrees of pedagogical potential. Findings suggest that many contemporary AI tools are structurally

designed to extend beyond procedural replacement, though such capacity does not guarantee transformative classroom practice. By reframing SAMR through an affordance lens, this study offers a systematic approach for evaluating AI tools based on pedagogical potential rather than assumed instructional impact.

Keywords: affordance theory, artificial intelligence, K-12 education, qualitative comparative analysis, SAMR model

Introduction

The integration of artificial intelligence into K-12 educational settings has accelerated dramatically in recent years, with tools ranging from intelligent tutoring systems to generative language models now widely accessible to teachers and students (Huang et al., 2023). While this rapid proliferation presents possibilities for personalized learning, adaptive instruction, and enhanced student engagement, it also raises critical questions about pedagogical transformation. As districts invest in AI technologies and educators experiment with new platforms, a fundamental challenge persists: distinguishing between tools that merely digitize existing practices and those that genuinely transform teaching and learning.

Multiple review studies reach a similar conclusion: while research on AI in K-12 is growing, explicit links between tool capabilities and established pedagogical frameworks remain limited. For example, Lee and Kwon's (2024) systematic review of K-12 AI education (2018–2023) catalogues topics and strategies but notes that most studies are small-sample, short-

duration classroom projects with limited evaluative depth, rather than theoretically framed integrations that would allow clear placement on a transformation continuum. In a focused area, Zhang and Tur's (2024) review of ChatGPT in K-12 highlights an early-stage literature dominated by practical use cases and ethical concerns, again without consistent alignment to instructional design models. Extending to language education, Wang et al., (2025) report fragmented theoretical lenses across generative-AI studies and call for clearer integration frameworks to guide curriculum and teacher practice.

Recent national surveys reveal this disconnect as well; while most K-12 teachers acknowledge the instructional potential of AI, fewer than one-third report active classroom use (Cheah et al., 2025), and one in four perceive AI tools as more harmful than helpful (Beshay, 2024). This disparity reveals an important paradox: while AI technologies promise to enhance differentiation, feedback, and student engagement, many educators remain hesitant to adopt them. Their caution often stems from ethical and pedagogical concerns, such as data privacy, academic integrity, algorithmic bias, and fears of deskilling the teaching profession (Hockly, 2023; Paglialunga & Melogno, 2025). Without clear guiding principles that show how AI can complement rather than replace teacher expertise, skepticism persists. Yet, this reluctance underscores precisely why structured analyses of the pedagogical value of AI are needed; if teachers perceive AI as harmful, understanding which tools genuinely transform learning becomes essential to building trust, efficacy, and equity in adoption. Without structured evaluation methods, educators and administrators struggle to make informed decisions about which tools warrant investment and how to integrate them meaningfully into curriculum and practice.

This lack of clarity has consequences for equity, teacher readiness, and student learning outcomes. When AI tools are implemented without pedagogical consideration, they risk exacerbating existing inequities or reinforcing traditional less-effective models of instruction rather than promoting innovation. This gap between technological availability and pedagogical understanding therefore represents more than an implementation issue; it reflects a broader need for clear frameworks to help educators assess whether AI tools enhance existing workflows or enable fundamentally new learning experiences. With approximately 60 percent of U.S. school districts now planning formal AI training for teachers, according to a 2025 national report from the RAND Corporation's American School District Panel (Diliberti et al., 2025), district leaders face a pivotal challenge: translating this rapid expansion of professional development into evidence-based, pedagogically meaningful implementation. At the policy level, the lack of shared frameworks could hinder cooperative decision-making and accountability for AI education. As districts expand AI initiatives, strategic planning and resource allocation benefit from evidence-based guidance that supports coherent, long-term implementation. At the same time, researchers play a vital role in establishing shared vocabularies and methods for examining the influence of AI on student agency, engagement, and achievement.

The purpose of this study is to systematically evaluate AI educational tools through the lens of the Substitution, Augmentation, Modification, Redefinition (SAMR) model (Puentedura, 2009, 2013), a widely recognized heuristic for assessing the degree to which technology transforms teaching and learning. By categorizing 26 widely-used AI tools across SAMR's four-level continuum from basic substitution to complete redefinition, this analysis identifies patterns in current K-12 AI implementation and reveals where pedagogical transformation is occurring versus where tools remain confined to enhancement functions. The study asks: (1) How do

current AI educational tools for K–12 settings align with the SAMR model when analyzed through their structural pedagogical affordances? and (2) What patterns emerge in the configurations of structural pedagogical affordances associated with these tools?

Using directed content analysis (Hsieh & Shannon, 2005), this research evaluates each tool based on publicly available documentation, feature descriptions, and demonstrated capabilities. Rather than focusing solely on technical specifications, the analysis emphasizes pedagogical affordances, including how tools support or reshape instructional design, learning tasks, and student agency. The resulting categorization offers practical guidance for educators while contributing to broader scholarly conversations about meaningful, equitable, and transformative AI integration in schools (Blundell et al., 2022; Krisbiantoro & Ashari, 2024).

In addition to content analysis, this study applies crisp-set Qualitative Comparative Analysis (Ragin, 1987) to examine how configurations of structural pedagogical affordances align with different degrees of task change described by the SAMR model. By examining patterns of co-occurrence across a structured set of pedagogical and functional conditions, QCA moves beyond individual tool evaluations to uncover the minimal configurations necessary for high-level instructional change. This approach offers a systematic lens for understanding how intersecting design features enable meaningful integration of AI in education. This approach complements the classification efforts of the content analysis with a configurational, evidence-based perspective on transformation.

The findings from this study hold relevance for multiple stakeholders. For classroom teachers, the SAMR-based analysis provides a replicable method for evaluating new tools and planning intentional integration. For administrators and instructional leaders, it offers a

structured analytic approach to guide professional development and AI-related policy decisions. For researchers, it addresses methodological gaps in AI education scholarship by operationalizing a systematic approach to tool evaluation. This study responds to the need for systematic evaluation approaches that help educators distinguish among the structural pedagogical affordances embedded in AI tools across the full SAMR continuum.

Literature Review

Theoretical Foundation of the SAMR Model

The Substitution, Augmentation, Modification, Redefinition (SAMR) model was introduced by Puentedura (2009) as a framework for evaluating the degree to which technology transforms teaching and learning. Initially conceptualized to support reflective technology integration, the SAMR model categorizes digital tools along a continuum from enhancement to transformation.

At the Substitution level, technology replaces a traditional task with no functional improvement. For example, typing an essay instead of handwriting it. Augmentation adds minor functional improvements, such as using digital editing or speech-to-text features that make the task more efficient. Modification represents a significant redesign of the task, where technology enables new processes or interactions, such as collaborative, cloud-based peer feedback. Redefinition occurs when technology enables experiences that would be impossible without it, such as AI-generated simulations, adaptive tutoring, or multimodal student compositions that merge human and machine creativity. Substitution and augmentation describe enhancements of existing pedagogical practices, while modification and redefinition denote the creation of new learning experiences previously inconceivable without technology (Puentedura, 2013).

Recent studies have extended and critiqued the model's application. Crompton (2020), through a systematic review of mobile learning in PK-12 classrooms, found that over half the studies aligned with SAMR's transformational levels, particularly in literacy and special education. However, Hamilton et al. (2016) argue that aligning higher SAMR levels with improved learning outcomes oversimplifies the complex interaction of learner, teacher, and context. Blundell (2022) echoes this concern in a scoping review, noting that SAMR applications often overlook contextual factors, particularly in digital assessment practices. More recent analyses of SAMR in AI-infused teacher education similarly emphasize contextual and ethical considerations. Kohnke & Zou (2025) integrated SAMR with TPACK to promote a critical, justice-oriented lens in TESOL preparation, underscoring that transformative technology use must be paired with ethical awareness. Although the SAMR model remains a widely used lens for examining technology integration, its application, particularly in emerging contexts such as AI, has been criticized for interpretive ambiguity and inconsistent alignment with pedagogical intent. This underscores the need for more systematic and pedagogically grounded classification of AI tools within such models.

Foundational perspectives in educational theory, including constructivism (Vygotsky, 1978; Bruner, 1996) and critical pedagogy (Freire, 1970), emphasize that learning occurs through interaction, reflection, and dialogue. These traditions position teachers and learners as co-constructors of knowledge rather than transmitters and recipients. When viewed through an affordance lens, these pedagogical traditions highlight how technology becomes meaningful not for its novelty, but for the action possibilities it makes available for dialogic, adaptive, and student-centered learning. For example, AI tools that personalize feedback, generate adaptive learning pathways, or facilitate metacognitive reflection can extend student agency,

collaboration, and problem-solving in ways consistent with constructivist and critical aims. Conversely, tools that merely automate grading or content delivery remain confined to enhancement levels of SAMR because they replicate existing routines rather than transform them. In this sense, the SAMR model provides a structural lens for describing how AI-mediated learning tasks may reflect varying degrees of procedural efficiency, functional enhancement, or task redesign consistent with dialogic and participatory pedagogies (Laurillard, 2013).

Further, Alivi (2019) underscores the model's utility in lesson planning and mobile learning design but cautions that its subjectivity may limit generalizability. State frameworks, such as the Louisiana Department of Education's (2024) *Framework for AI Integration*, explicitly map AI use to SAMR's four levels, illustrating how substitution-to-redefinition continua are now being operationalized in K-12 guidance. Krisbiantoro (2024), in a systematic review of university-level English language teaching, found that educators often remain at the substitution level due to barriers such as insufficient training and institutional resistance. These findings point to the importance of situating SAMR within broader discussions of teacher preparation, digital equity, and contextual adaptability.

While SAMR provides a structural continuum for describing technology integration, its interpretive flexibility invites further theoretical grounding. One approach to strengthening analytic clarity is through the lens of affordances, which connects technological features to pedagogical action possibilities.

Current State of AI in Education

The integration of AI into educational settings has rapidly expanded, offering both opportunities and challenges across disciplines and grade levels. In K-12 contexts, AI tools now

support a wide range of instructional, assessment, and administrative functions, from intelligent tutoring systems to predictive analytics and generative learning companions. These tools leverage natural language processing, automated speech recognition, machine learning, and large language models (LLMs) to deliver adaptive, personalized, and scalable educational experiences (Huang, 2023).

A significant portion of artificial intelligence in education research centers on language learning, where chatbots, virtual tutors, and intelligent computer-assisted language learning systems are increasingly prominent. Huang (2023) found that applications such as grammar correction, speaking practice in virtual reality, personalized learning recommendations, and real-time feedback have grown steadily in both classroom use and scholarly attention. These developments reflect how AI supports second-language acquisition through adaptive scaffolding and interactive immersion, aligning with pedagogical trends that emphasize autonomy and feedback-driven learning. Similarly, Thomas et al. (2024) emphasizes that LLMs extend the capabilities of adaptive feedback systems by generating on-demand content, guidance, and prompts tailored to individual learner input, demonstrating how AI can mediate dynamic, conversational forms of language practice.

Beyond language education, AI tools have been applied across diverse domains, including mathematics instruction, social studies engagement, and emotional support for students with learning disabilities (Paglialunga & Melogno, 2025). These cross-disciplinary applications illustrate the growing versatility of AI as a pedagogical partner rather than a subject-specific tool. Paglialunga and Melogno's (2025) multiple-strategy research found that adaptive platforms such as DreamBox and Lexia supported both academic growth and emotional well-being, particularly for students with dyslexia and dyscalculia, highlighting the potential of AI to promote inclusion

through personalized, responsive feedback. Together, these strands of research underscore the dual capacity of AI to enhance domain-specific instruction and advance educational equity through individualized learning experiences.

Despite these advancements, concerns persist across the educational landscape. For teachers, apprehension often centers on instructional design and professional agency. Luckin (2019) argues that AI tools are often developed without adequate input from educators or insights from learning sciences, leading to misaligned instructional designs. This disconnect contributes to uncertainty about how AI can meaningfully support teaching rather than replace or oversimplify it. Empirical findings mirror national trends: Cheah et al. (2025) found that while most K-12 teachers acknowledge AI's instructional potential, fewer than one-third report active classroom use. National survey data corroborate this hesitation, with one in four U.S. teachers perceiving AI tools as more harmful than helpful (Beshay, 2024).

Students and schools also face distinct challenges. Generative AI systems such as ChatGPT raise questions about authenticity, authorship, and bias, especially when used without oversight (Paglialunga & Melogno, 2025). Ethical concerns around data privacy, algorithmic transparency, and equitable access remain salient, particularly in under-resourced schools (Hockly, 2023). Collectively, these concerns underscore the need for theoretically grounded analytic approaches that clarify how AI tools' structural affordances align with pedagogical models in K-12 education.

The current state of AI in education reflects a dynamic and evolving field marked by innovation and optimism, alongside legitimate pedagogical and ethical concerns. While the capabilities of AI tools continue to expand, their effective integration into K-12 practice requires

not only technological sophistication but also pedagogical intentionality, professional development, and ethical governance as evidenced with Diliberti et al. (2025) reporting that 60 percent of districts planned formal AI training, indicating a systemic shift toward structured implementation.

Technology Integration Frameworks in Education

To evaluate the role of digital tools in teaching and learning, researchers and educators have adopted a range of technology integration models. Among the most widely cited are the SAMR model (Puentedura, 2009, 2013), Technological Pedagogical Content Knowledge (TPACK) framework (Mishra & Koehler, 2006), and the Technology Integration Matrix (TIM) (Florida Center for Instructional Technology, 2011). Each model offers a distinct lens for examining how technology mediates pedagogical practice and shapes learning outcomes.

The SAMR model has been widely used in studies of mobile learning and instructional design (Crompton, 2020) and is often praised for its simplicity and accessibility (Alivi, 2019). However, critics argue that its hierarchical structure and interpretive flexibility can obscure the role of context, pedagogy, and learner needs (Hamilton et al., 2016). These critiques highlight the need for theoretically grounded analytic approaches that connect tool design features to pedagogical action possibilities. Affordance theory offers one such lens by linking technological capabilities to instructional potential.

The TPACK framework, by contrast, emphasizes the dynamic interaction between technological, pedagogical, and content knowledge. It recognizes that effective technology use is inseparable from the subject matter and teaching approach, offering a more nuanced understanding of teacher decision-making. Some scholars, such as Krisbiantoro (2024), have found that SAMR and TPACK can be complementary; SAMR provides a structural

classification, while TPACK foregrounds teacher expertise and knowledge integration in tool selection and implementation. Recent work supports this intersection of frameworks in AI-integrated teacher education. Kohnke & Zou (2025) demonstrated how SAMR and TPACK can operate synergistically when guiding educators' use of generative AI tools, particularly in TESOL contexts. Their study highlights that aligning the transformational stages of SAMR with knowledge domains of TPACK encourages both pedagogical depth and ethical reflection in technology integration. This synthesis underscores how AI adoption can move beyond substitution toward contextually grounded, justice-oriented transformation.

The TIM further extends these perspectives by cross-referencing five levels of technology integration (Entry, Adoption, Adaptation, Infusion, Transformation) with five characteristics of learning environments (Active, Collaborative, Constructive, Authentic, Goal-Directed). This matrix allows for a multidimensional evaluation of classroom technology use, accounting for both pedagogical intent and student engagement. As it provides richer detail for evaluating authentic instructional practices across disciplines and settings, TIM is more complex than SAMR.

Recent literature also references emerging models such as based on the Passive, Interactive, Creative $\times$ Replacement, Amplification, Transformation (PICRAT) model, which explicitly considers student engagement alongside instructional impact (Kimmons, 2020).

Though not yet widely applied in empirical AI education studies, PICRAT has been proposed as a useful alternative when analyzing generative AI and student-centered tool usage (Grossmann, 2023).

Despite these models' contributions, a persistent gap remains: few studies systematically map AI tools across these frameworks or evaluate how specific features support movement along

their integration continua. As Blundell (2022) observes in their scoping review of SAMR applications, there is a lack of alignment between research findings and practical classification of technology-enhanced learning. The current study addresses this gap by employing SAMR to systematically evaluate a range of AI educational tools and identify their transformative potential in K-12 settings.

Gap in the Literature Regarding Systematic Analysis of AI Educational Tools

Much of the existing literature focuses on either the theoretical potential of AI in education (Luckin, 2019) or the functionality of specific tools such as chatbots, adaptive platforms, or LLM-powered applications (Huang, 2023; Kohnke, 2023). Despite growing scholarship on AI in education, limited research systematically connects tool design features to models of instructional change in ways that distinguish structural pedagogical potential across enhancement- and transformation-oriented levels.

Several reviews highlight this fragmentation. Crompton's (2020) systematic review of mobile learning through the SAMR model underscores a promising trend toward transformative applications, but they do not examine AI tools specifically. Blundell (2022) similarly notes inconsistencies in how SAMR levels are applied in educational research and calls for more rigorous alignment between tool features and pedagogical categorization. As Zhang and Tur (2024) noted in a systematic review, most studies remain exploratory and lack clear pedagogical classification, reinforcing the need for SAMR-based analyses. Meanwhile, Krisbiantoro (2024) identifies recurring substitution-oriented uses of technology, particularly in language learning contexts, highlighting barriers such as limited training and institutional resistance that shape how AI tools are enacted in practice.

In parallel, studies of AI integration often emphasize innovation without grounding tool evaluation in pedagogical frameworks. For example, tools such as ChatGPT and adaptive feedback systems are lauded for their personalization capabilities (Paglialunga and Melogno, 2025), yet few analyses assess whether they reconfigure learning tasks or student agency in ways consistent with transformative models like SAMR or TPACK. Instead, research tends to remain descriptive and focused on tool functionality, user experience, or short-term outcomes without linking design affordances to broader educational goals (Ghosh, 2023; Su, 2020). Recent meta-analytic findings (Deng et al., 2025) demonstrate statistically significant learning gains from ChatGPT-mediated instruction, yet few studies systematically connect AI tool design features to pedagogical affordances and structured models of instructional change.

Moreover, the diversity of AI tool types, ranging from intelligent tutoring systems to predictive modeling platforms, has made systematic comparison difficult. Without a shared analytic approach, educators and researchers may struggle to distinguish among tools that support procedural efficiency, functional enhancement, or more extensive task redesign within specific instructional contexts.

Despite the rapid expansion of artificial intelligence tools in K-12 education, systematic approaches that connect tool design features to pedagogical meaning remain limited. Much of the existing literature describes functionality, reports exploratory classroom implementations, or discusses ethical considerations without analytically linking structural tool capabilities to models of instructional change. As a result, educators and researchers lack systematic analytic approaches for distinguishing among tools that support procedural efficiency, functional enhancement, or more extensive task redesign. Addressing this gap requires an analytic approach

grounded in affordance theory that connects technical features to pedagogical action possibilities in transparent and theoretically grounded ways.

Theoretical Framework: SAMR Through an Affordance Lens

This study employs the Substitution, Augmentation, Modification, Redefinition (SAMR) model (Puentedura, 2009, 2013) as a structural lens for examining the pedagogical potential of AI-powered educational tools in K-12 contexts. The SAMR model conceptualizes technology integration along a continuum distinguishing enhancement-oriented task modification (Substitution and Augmentation) from transformation-oriented task redesign (Modification and Redefinition). While SAMR has been widely adopted in both research and practice due to its accessibility and clarity, it has also been critiqued for its perceived hierarchical structure and reliance on subjective interpretation (Hamilton et al., 2016). Critics caution against equating higher SAMR levels with inherently superior learning outcomes, noting that instructional quality depends on context, task design, and pedagogical intent rather than categorical placement alone.

To address these concerns and strengthen the analytic grounding of SAMR classifications, this study integrates affordance theory as an analytic bridge between tool features and pedagogical interpretation. Originating in ecological psychology (Gibson, 1979), affordances refer to the action of possibilities made available by an object or environment relative to a user. Norman (1988) extended this concept to design contexts, emphasizing that affordances are relational; they emerge through interaction between users, tools, and tasks rather than existing as fixed properties of the technology itself. In educational settings, affordances describe the instructional possibilities that tools make available for teaching and learning (Laurillard, 2013).

Applying an affordance lens shifts the focus from cataloging technical features to examining how those features expand or constrain pedagogical action. For example, content generation capabilities may afford rapid prototyping of instructional materials; natural language processing may afford dialogic feedback; personalization features may afford differentiated pathways; and real-time analytics may afford adaptive instructional decision-making. These affordances represent structural pedagogical possibilities embedded in tool design configurations. Importantly, they reflect potential rather than enacted classroom outcomes.

Within this study, SAMR levels are therefore interpreted not as hierarchical indicators of quality but as configurations of pedagogical affordances associated with varying degrees of task change. Substitution reflects tools whose affordances primarily replicate existing tasks with minimal expansion of instructional action. Augmentation reflects tools that afford functional improvements such as efficiency, feedback, or refinement while maintaining the original task structure. Modification reflects affordances that enable meaningful task redesign, such as iterative feedback loops, dynamic scaffolding, or restructured collaboration. Redefinition reflects affordances that enable task architectures or learning experiences that would be difficult to enact without the technology.

By integrating affordance theory, this study reframes SAMR categorization as an analysis of pedagogical potential grounded in structural design features rather than as evidence of realized classroom transformation. This distinction is critical. A tool's placement within a SAMR category does not assert that transformation has occurred in practice; rather, it indicates that the tool's affordance configuration enables particular forms of instructional action under appropriate

pedagogical conditions. In this way, all SAMR levels carry distinct forms of pedagogical potential depending on instructional goals, learner needs, and contextual constraints.

This theoretical integration strengthens the systematic evaluation of AI tools' structural pedagogical affordances by anchoring classification decisions in identifiable affordance patterns while maintaining sensitivity to the contextual and relational nature of teaching and learning. It also provides a coherent foundation for the subsequent qualitative content analysis and configurational modeling conducted in this study.

Research Questions

The purpose of this study is to systematically examine how AI-powered educational tools used in K-12 settings align with the SAMR model when analyzed through their structural pedagogical affordances. Specifically, the study seeks to (1) classify tools according to degrees of SAMR-aligned task change based on their affordance configurations, and (2) identify patterns in the combinations of affordances associated with different levels of task redesign.

1. How do current AI educational tools for K-12 settings align with the SAMR model when analyzed through their structural pedagogical affordances?
2. What configurations of structural pedagogical affordances are associated with different degrees of SAMR-aligned task change?

Methods

Design

The study employed Directed Content Analysis (Hsieh & Shannon, 2005) to systematically evaluate the structural pedagogical affordances of AI educational tools in K-12

contexts. This method complements the use of the SAMR model (Puentedura, 2009, 2013) in guiding coding decisions and classifying how AI tools' structural affordances align with degrees of task change. To further examine patterns in how affordances combine across tools, the study employed crisp-set Qualitative Comparative Analysis (csQCA) (Ragin, 1987) to identify configurations of structural pedagogical affordances associated with higher degrees of SAMR-aligned task change. Together, these methods provide a replicable analytic approach for evaluating AI tools' structural pedagogical potential in K-12 education.

The units of analysis were individual AI educational tools identified as relevant for K-12 educational contexts. An initial list of 39 AI tools was compiled through systematic searches of widely recognized educational technology sources, including EdSurge, Common Sense Education, *Education Week* research reports, and Edutopia. Tool lists from each source were compiled and cross-referenced to identify recurring and widely discussed platforms. Tools appearing across multiple sources, along with publicly available information regarding intended user populations, were prioritized for inclusion. A matrix was developed to determine whether each tool met the inclusion criteria outlined in Table 2. Only tools receiving a score of 1.0 across all four inclusion criteria were retained for analysis, resulting in a final sample of 26 AI tools. A threshold of 1.0 was selected to ensure that included tools demonstrated clear relevance to K-12 contexts, functional diversity, identifiable pedagogical applications, and educational visibility. The resulting sample included a range of AI tool categories commonly discussed in K-12 contexts, including generative chat systems, integrated workflow tools, and adaptive platforms.

Table 1
AI Tools Selection Matrix

K12_RE										
Tool	L	FUNC	DIV	PED	POT	ED	VIS	GEN	EDU INC	SCORE

Beautiful AI	1	0	0	1	GEN	0.5
Brisk Teaching	1	1	1	1	EDU	1
Canva	0	0	1	1	GEN	0.5
ChatGPT	1	1	1	1	GEN	1
Classcraft	1	1	1	0	EDU	0.75
Classpoint AI	1	1	1	1	EDU	1
Claude	1	1	1	1	GEN	1
CoPilot	0	1	1	1	GEN	0.75
Curipod	1	1	1	1	EDU	1
Diffit	1	0	0	1	EDU	0.5
Easybib	1	0	1	1	EDU	0.75
Edcafe AI	1	1	1	0	EDU	0.75
Eduaide AI	1	1	1	1	EDU	1
Figma	1	1	1	1	GEN	1
Formative	1	1	1	1	EDU	1
Gamma	1	1	1	1	GEN	1
Gemini	1	1	1	1	GEN	1
Gradescope	1	1	1	1	EDU	1
Grammarly	1	1	1	1	GEN	1
Kahoot	1	1	1	1	EDU	1
Khanmigo	1	1	1	1	EDU	1
Knowit	1	1	1	0	EDU	0.75
Learning Genie	1	1	1	0	EDU	0.75
Magic School AI	1	1	1	1	EDU	1
Nearpod	1	1	1	1	EDU	1
NotebookLM	1	1	1	1	GEN	1
OtterAI	1	1	1	1	GEN	1
Owlfit	1	1	1	1	EDU	1
Padlet	1	1	1	1	EDU	1
Pear Deck	1	0	1	0	EDU	0.5
Perplexity AI	1	1	1	1	GEN	1
Quillbot	1	0	1	1	GEN	0.75
Quizizz	1	1	1	1	EDU	1
Quizlet	1	1	1	1	EDU	1
School AI	1	1	1	0	EDU	0.75
Snorkl	1	0	1	0	EDU	0.5
Squirrel AI	1	1	1	1	EDU	1
TeacherServer	1	1	1	1	EDU	1

Turnitin	1	1	1	1	EDU	1
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Note- INC_SCORE = Inclusion score, calculated as the mean of K12_REL (K–12 relevance), FUNC_DIV (functional diversity), INST_FUNC (instructional functionality), and ED_VIS (educational visibility). GEN_EDU indicates whether the tool is general-purpose (GEN) or education-specific (EDU). Tools with an inclusion score of 1.0 were retained for the final analytic sample (n = 26).

Tools that did not receive a score of 1.0 across all four inclusion criteria were excluded from the final analytic sample. Excluded tools typically did not meet one or more of the following criteria: demonstrated relevance to K-12 contexts, functional diversity across instructional tasks, identifiable pedagogical applications, or current educational visibility. For example, tools such as Beautiful AI and Diffit did not meet all inclusion criteria related to pedagogical scope or K-12 specificity. Others, including Canva, Quillbot, and Easybib, were excluded due to limited AI-driven functionality relative to the defined inclusion standards. Tools such as Classcraft, Edcafe AI, School AI, and Knowit did not meet the established threshold for educational visibility or current K-12 adoption. The final analytic sample of 26 tools was determined strictly based on the inclusion scoring matrix presented in Table 1, with selection criteria operationalized as detailed in Table 2.

Table 2

AI Tool Selection Criteria

Criteria	Code	Indicator	Coding
Relevance to K-12 Contexts	K12_REL	Tool is currently used in K–12 classrooms, supports grade-level use, or explicitly targets K–12 teaching	1 = Yes, 0 = No
Functional Diversity	FUNC_DIV	Tool supports multiple instructional or classroom functions (e.g., lesson planning, assessment, student engagement, instructional management)	1 = Yes, 0 = No
Instructional Functionality	INST_FUNC	Tool offers instructional customization, scaffolding, adaptive pathways, or learning	1 = Yes, 0 = No

Educational Adoption/Visibility	ED_VIS	design features Tool demonstrates documented visibility in professional education publications, educator communities, or established educational technology platforms	1 = Yes, 0 = No
General vs Education-Specific AI Mix	GEN_EDU	Includes tools made specifically for education and general AI tools adapted for learning use cases	EDU = Education-specific, GEN = General purpose

Note- GEN_EDU classification was not included in the inclusion score calculation.

By categorizing 26 widely used AI tools across the SAMR model's four levels of task change, this analysis identifies patterns in how tools' structural pedagogical affordances align with enhancement- and transformation-oriented task design.

The study employed Hsieh and Shannon's (2005) approach to Directed Content Analysis. Consistent with this approach, coding was guided by a priori theoretical framework integrating the SAMR model and affordance theory. Structural pedagogical affordances were systematically coded according to their alignment with defined SAMR categories. Operational definitions were developed for each SAMR level and sublevel, and a comprehensive codebook was constructed to articulate indicators, inclusion criteria, and decision rules (see Table 3).

Table 3
Coding Rules for SAMR Levels

SAMR Level	Coding Rules
Substitution	S1 Tool affordances replicate an existing task without altering task structure. S2 Tool affordances provide limited functional enhancement (e.g., efficiency or formatting improvements) while maintaining the original task structure.
Augmentation	A1 Tool affordances introduce functional enhancements that support feedback, refinement, or minor interaction shifts

	without restructuring the task.
	A2 Tool affordances meaningfully expand task functionality (e.g., real-time feedback, adaptive support) while preserving the core task design.
Modification	M1 Tool affordances enable partial task redesign, altering workflow, interaction patterns, or instructional sequencing. M2 Tool affordances enable substantial task redesign that reconfigures how learners engage with content, peers, or feedback.
Redefinition	R1 Tool affordances enable task structures that would be difficult to implement without the technology. R2 Tool affordances enable task architectures that are structurally dependent on the technology and substantially alter instructional possibilities.

Note- These coding rules reflect structural pedagogical affordances embedded in tool design rather than observed classroom implementation.

Following development of the coding rules, researchers conducted systematic feature documentation for each tool. Each platform was independently reviewed using publicly available sources, including official websites, feature descriptions, instructional videos, and tutorials. Structural pedagogical affordances were documented according to predefined codebook indicators and evaluated based on described functionality rather than observed classroom use. As such, classifications reflect documented design features rather than enacted instructional practice. Because data were derived from publicly available platform documentation and demonstrations, classifications reflect presented functionality, which may emphasize innovative features. While efforts were made to apply consistent coding criteria, the analysis is limited to the affordances described in official materials rather than independently verified classroom implementation.

Directed Content Analysis

Consistent with Hsieh and Shannon's (2005) approach to Directed Content Analysis, researchers applied a priori coding categories derived from the integrated SAMR–affordance framework. Each tool was classified according to predefined SAMR sublevels based on documented structural pedagogical affordances. Coding decisions were grounded in operational definitions articulated in the codebook, emphasizing the degree of task change structurally enabled rather than the presence of isolated technical features.

To ensure coding validity, intercoder reliability was conducted using Cohen's Kappa. Two researchers independently coded all tools prior to calculating interrater agreement. Interrater reliability was assessed to evaluate consistency between the two raters' SAMR classifications of the 26 AI educational tools. Using ReCal2 (Freelon, 2010), percent agreement between raters was 61.5%, with Cohen's $\kappa = .42$. According to Landis and Koch (1977), this level of agreement falls within the moderate range. Moderate agreement is consistent with prior scholarship noting interpretive flexibility in SAMR applications (Hamilton et al., 2016) and reflects the complexity of classifying multifunctional AI tools within discrete categorical levels.

Initial discrepancies between raters primarily emerged in cases where tools exhibited multiple overlapping affordances that could plausibly align with more than one SAMR level. Because many AI systems simultaneously support generative, adaptive, and feedback-oriented functions, distinguishing adjacent SAMR levels required careful attention to the degree of task change structurally enabled rather than the presence of advanced features alone.

To strengthen analytic rigor, the two raters engaged in a structured adjudication process following the initial independent coding. Disagreements were systematically reviewed, decision rules were clarified in the codebook, and each disputed case was re-evaluated considering the affordance-based criteria outlined in the theoretical framework. Through this structured

consensus process, full agreement was reached on all cases prior to subsequent configurational analysis.

The finalized consensus-coded dataset was used for the crisp-set Qualitative Comparative Analysis (csQCA). Reporting both initial reliability statistics and the adjudication process enhances transparency and ensures that the final dataset reflects a unified, theoretically grounded classification of structural pedagogical potential.

Table 4
Interrater Reliability for SAMR Model Classifications (n = 26 Tools)

Statistic	Value
Percent agreement	61.5%
Cohen's κ	.42
Scott's π	.41
Krippendorff's α (nominal)	.42
N Agreements	16
N Disagreements	10
N Cases	26
N Decisions	52

Note. Initial reliability statistics reflect independent coding prior to consensus adjudication. Final SAMR classifications used in subsequent analyses reflect negotiated agreement between raters.

Following consensus coding, researchers conducted exploratory pattern analysis to examine the distribution of tools across SAMR levels and to identify recurring configurations of structural pedagogical affordances. Frequency distributions were calculated to assess the prevalence of tools aligned with varying degrees of SAMR-defined task change. In addition, co-occurrence patterns between affordance indicators and SAMR classifications were examined to

identify structural clustering across levels. This preliminary analysis informed the subsequent configurational modeling using crisp-set Qualitative Comparative Analysis (csQCA).

Qualitative Comparative Analysis

To address the research question examining configurations of structural pedagogical affordances across AI-powered tools in K-12 settings, the study employed Ragin's (1987) crisp-set Qualitative Comparative Analysis (csQCA). This approach was used to examine how combinations of affordance indicators aligned with transformation-oriented SAMR classifications.

Tools were assigned a binary outcome score based on whether their highest SAMR classification aligned with transformation-oriented task change, defined as Modification-level M2 or any Redefinition-level classification. A score of 1 indicated alignment with transformation-oriented task redesign, whereas 0 indicated alignment with enhancement-oriented classifications only.

The outcome condition represented transformation-oriented SAMR alignment, and explanatory conditions represented structural pedagogical affordances derived from the content analysis. Each of the 26 tools was coded for the presence (1) or absence (0) of affordance indicators derived from tool design features. Conditions included generative affordances (CG), adaptive affordances (PERS), dialogic affordances (NLP), real-time feedback affordances (RTF), and related structural capabilities. These indicators were operationalized dichotomously to allow systematic examination of configurational pathways associated with differing degrees of SAMR-aligned task change.

A truth table was constructed using the QCA package in R (Dusa, 2019), representing empirically observed configurations of conditions across cases. Only configurations with

empirical instances were retained. To balance analytic depth with interpretive parsimony and maintain an appropriate condition-to-case ratio, three model specifications were examined. The primary analysis focused on an 8-condition model representing core pedagogical affordance clusters identified as theoretically central: content generation (CG), student-centered creation (STU), natural language processing (NLP), real-time feedback (RTF), instructional insight (INSIGHT), personalization (PERS), workflow integration (WORK), and assessment automation (AA).

A 10-condition intermediate model and a 14-condition full model were subsequently analyzed as robustness checks. Given the relatively small number of cases ($n = 26$), the reduced 8-condition model served as the primary analytic specification to reduce the risk of overfitting and excessive logical remainders while maintaining theoretical coherence.

Findings

Alignment with each SAMR Phase

Analysis of the 26 AI educational tools revealed distribution patterns across the SAMR continuum based on structural pedagogical affordances. Specifically, 42% of the tools ($n = 11$) demonstrated affordance configurations consistent with Redefinition-level task change, indicating structural capacity for task designs that would be difficult to enact without the technology. Another 42% ($n = 11$) aligned with the Modification level, reflecting affordances associated with substantial task redesign. In contrast, 15% ($n = 4$) aligned with Augmentation-level classifications, reflecting functional enhancements that preserve core task structure while improving efficiency or feedback mechanisms. No tools were classified as purely Substitution-

level based on their highest affordance configuration. This pattern suggests that tools meeting the inclusion criteria tended to incorporate multiple structural affordances extending beyond direct task replication. Figure 1 displays the distribution of tools across SAMR levels according to their highest coded SAMR classification.

Figure 1
Distribution of Tools Across SAMR Levels



The absence of tools classified as purely Substitution-level warrants clarification. This finding does not suggest that AI tools are never used for basic task replacement in practice.

Rather, it reflects the structural design characteristics of the tools included in this analysis. Many contemporary AI systems embed generative, adaptive, or feedback-oriented affordances by default. As a result, even when used in substitution-oriented ways, their structural design typically includes capabilities that extend beyond direct task replacement. In this sense, the

absence of “pure” Substitution classifications likely reflects the structural affordance configurations embedded in contemporary AI tool design rather than assumptions about classroom practice. However, this structural potential does not imply that classroom implementation will necessarily move beyond substitution-level use, as pedagogical enactment remains dependent on teacher intent, context, and instructional design.

At the Modification and Redefinition levels, tools such as Magic School AI, Notebook LM, and Brisk Teaching were associated with transformation-oriented affordance configurations through dynamic features like real-time feedback, adaptive pathways, and structured generative capabilities. These findings align with prior literature noting that AI systems incorporating natural language processing can support individualized feedback, interactive dialogue, and expanded instructional design options (Mejeh & Rehm, 2024; Puentedura, 2013).

Although no tools were classified as purely Substitution-level in terms of their highest affordance configuration, some tools demonstrated substitution-oriented affordances within narrower use cases. For example, Turnitin and Classpoint AI may exhibit affordances consistent with task replication in narrower instructional use cases without fundamentally altering pedagogical structure. Augmentation-level tools, including Perplexity AI and Grammarly, offered expanded functional affordances. These applications enhanced feedback loops and differentiated instruction but remained teacher-directed, aligning with the A1 and A2 subcategories of the coding scheme.

Although this analysis focuses on structural affordance configurations rather than enacted use, the distribution of classifications aligns with national reports indicating that AI tools are frequently adopted for productivity-oriented functions in classroom contexts (Diliberti et al.,

2025). This contrast between structural capacity and typical classroom usage underscores the distinction between technological design and pedagogical enactment.

Configurational Analysis of Affordance Pathways

The primary configurational analysis focused on an 8-condition model representing core pedagogical affordances identified as theoretically central: content generation (CG), student-centered creation (STU), natural language processing (NLP), real-time feedback (RTF), instructional insight (INSIGHT), personalization (PERS), workflow integration (WORK), and assessment automation (AA). This reduced model identified multiple pathways sufficient for alignment with transformation-oriented SAMR classifications. These pathways consistently included combinations of generative, dialogic, adaptive, and feedback-oriented affordances, suggesting that a limited set of structural capabilities is consistently associated with transformation-oriented task redesign. These pathways represent structural configurations of affordances rather than deterministic causal models of classroom transformation.

Extended analyses using 14-condition specifications yielded comparable patterns. While the full model produced additional configurational pathways, the core affordance combinations remained stable across specifications. Coverage values decreased slightly (0.909 and 0.727, respectively), supporting the interpretive parsimony of the reduced model while confirming that broader specifications did not substantially alter the primary configurational patterns.

Table 5

Truth Table Minimization Results for the 8-Condition Core Pedagogical Affordance Model

		inclS	PRI	covS	cases
1	CG*PERS*RTF*~AA*~WORK*INSIGHT*STU	1.000	1.000	0.182	12;19
2	CG*NLP*RTF*~AA*~WORK*~INSIGHT*STU	1.000	1.000	0.182	15;14

3	CG*~PERS**NLP*~RTF*AA*WORK*INSIGHT*STU	1.000	1.000	0.091	3
4	CG*PERS**~NLP*~RTF*AA*~WORK*~INSIGHT*STU	1.000	1.000	0.091	11
5	CG*PERS**NLP*RTF*AA*WORK*~INSIGHT*STU	1.000	1.000	0.091	9
6	~CG*~PERS**NLP*RTF*~AA*WORK*INSIGHT*~STU	1.000	1.000	0.091	4
M1		1.000	1.000	0.727	

Note- Each configuration of conditions details the presence or absence of each condition where a ~ indicates the absence of a condition and a * indicates the condition in conjunction with what follows in the configuration.

Across model specifications, several pedagogical affordances consistently emerged as central within transformation-oriented SAMR configurations. In both the reduced 8-condition model and extended specifications, content generation (CG), student-centered creation (STU), natural language processing (NLP), real-time feedback (RTF), and instructional insight (INSIGHT) appeared most frequently across sufficient solution terms. These recurring affordance clusters suggest that generative, dialogic, and feedback-oriented capabilities are structurally associated with transformation-aligned task redesign within the analytic framework employed in this study.

In contrast, collaboration features (COLLAB) and security provisions (SEC) appeared less consistently across model specifications, indicating that while important for implementation and governance considerations, they were less central within the transformation-oriented configurations identified through this analysis. The 10- and 14-condition models yielded comparable core patterns, though coverage values decreased slightly (0.909 and 0.727, respectively). This pattern reinforces the interpretive parsimony of the reduced model while indicating that broader specifications did not substantially alter the primary configurational findings.

Discussion

The findings reveal important structural patterns in the pedagogical potential of AI tools currently used in K-12 education. Rather than documenting enacted classroom transformation, this study examined the affordance configurations embedded within tool design and their alignment with degrees of task change described by the SAMR model. In this study, transformation-oriented classifications refer specifically to tools coded at M2 or Redefinition levels. The distribution of tools across Modification and Redefinition classifications reflects the structural capacity of many contemporary AI systems to support task redesign under appropriate instructional conditions.

At the same time, the presence of transformation-oriented affordances does not guarantee transformative classroom practice. Consistent with prior research (Cheah et al., 2025; Krisbiantoro, 2024), classroom enactment remains contingent upon teacher expertise, instructional goals, contextual constraints, and professional development support. As such, the findings should be interpreted as identifying structural pedagogical potential rather than realized instructional impact.

The results of the csQCA provide insight into how specific combinations of structural affordances align with transformation-oriented SAMR classifications in K-12 contexts. Notably, no single affordance was sufficient in isolation to align with transformation-oriented classifications. Instead, transformation-aligned outcomes were associated with intersecting capabilities that collectively reconfigure task structure within the SAMR model.

The most frequent conditions across configurations aligned with transformation-oriented classifications included content generation, student-centered creation, natural language processing, real-time feedback, and instructional insight. These recurring clusters suggest that tools incorporating generative, dialogic, and feedback-oriented affordances are structurally positioned to support task redesign within the SAMR model. Collectively, these features expand the range of instructional actions available to educators, allowing for increased interaction with content, differentiated feedback pathways, and iterative learning processes.

In contrast, collaboration features and security functions appeared less frequently across transformation-oriented configurations. While these capabilities remain important for usability, governance, and compliance considerations, they were less central within the structural configurations associated with task redesign in this analysis. This distinction suggests that within the SAMR model, transformation-aligned classifications are more directly connected to affordances that alter task structure rather than those that primarily support administrative or logistical processes.

Limitations

This study is limited to the analysis of structural pedagogical affordances embedded in AI tool design rather than enacted classroom practice. Accordingly, the findings reflect potential alignment with SAMR-defined degrees of task change and do not indicate how tools are implemented in real-world instructional contexts. Classroom enactment remains contingent upon teacher decision-making, instructional goals, and contextual factors not examined in this analysis.

Additionally, because data were derived from publicly available platform documentation, feature descriptions, and demonstrations, classifications reflect presented functionality rather than independently verified use. Platform materials may emphasize innovative or aspirational capabilities for marketing purposes, introducing potential promotional bias in how affordances are represented. While consistent coding procedures were applied, the analysis remains limited to documented structural features rather than observational or usage-based data.

The study is further constrained by its sample size ($n = 26$), which necessitated careful attention to condition-to-case ratios in the configurational analysis. Although reduced-model specifications were employed to enhance interpretive parsimony, the limited number of cases restricts the generalizability of identified configurations. In addition, the use of crisp-set QCA required dichotomous coding of affordances, which simplifies complex feature variation into binary presence or absence. While this approach enables systematic configurational modeling, it may not capture gradations in feature depth or quality across tools. Finally, QCA identifies patterns of configurational association rather than causal relationships, and findings should be interpreted accordingly.

Implication and Future Research

The findings from this study offer practical implications for multiple educational stakeholders. For classroom teachers, the affordance-informed SAMR analysis provides a structured approach for examining how AI tools align with varying degrees of task change. For administrators and instructional leaders, the model offers a systematic lens for guiding professional learning and technology adoption decisions. For researchers, the study contributes a methodological model for operationalizing structural feature analysis in AI education

scholarship. As AI technologies continue to evolve, analytically grounded frameworks can support more transparent evaluation of how tool design aligns with instructional goals.

Future research should extend beyond structural analysis to examine how educators and students engage with AI tools in classroom practice. Studies exploring how affordance-rich tools are enacted across diverse K-12 contexts would help clarify the relationship between structural capacity and instructional implementation. Additional research might also examine AI tools through complementary theoretical lenses, such as TPACK or Universal Design for Learning (UDL), to compare how different frameworks interpret tool-pedagogy alignment.

Developers and decision makers may also draw on the identified configurational patterns to inform design and selection processes. Understanding which combinations of structural affordances are associated with varying degrees of SAMR-aligned task change can support more deliberate alignment between tool features and instructional objectives.

Conclusion

This study examined how structural affordances embedded within widely used AI tools align with degrees of task change described by the SAMR model. Rather than asserting realized transformation in practice, the findings identify recurring affordance clusters associated with transformation-oriented SAMR classifications. These patterns indicate that generative, dialogic, and adaptive capabilities are structurally positioned to support task redesign under appropriate instructional conditions. At the same time, the analysis underscores that structural capacity alone does not determine classroom impact; instructional enactment remains dependent on pedagogical intent, teacher agency, and contextual factors.

This study reinforces the need for analytically grounded evaluation frameworks that connect technological design features to pedagogical interpretation (Deng et al., 2025; Kohnke et al., 2025; Mejeh & Rehm, 2024). By integrating affordance theory with the SAMR model, the study demonstrates a systematic approach for examining how AI tool design aligns with varying degrees of task change. Future research should extend this structural analysis to investigate classroom enactment, longitudinal implementation patterns, and the role of teacher professional learning in mediating affordance realization. Comparative analyses across frameworks such as SAMR, TPACK, and related policy models may further clarify how AI tools are interpreted and integrated within diverse educational contexts.

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Appendix A-

Truth Table Minimization Results for the 14-Condition Full Model

		inclS	PRI	covS	cases
1	CG*~PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*~INSIGHT*STU*~COLLAB*~ADMIN*~SEC	1.000	1.000	0.091	15
2	CG*PERS*~NLP*~RTF*AA*~MULTI*DOCX*LANG*~ WORK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	11
3	CG*PERS*~NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*INSIGHT*STU*COLLAB*~ADMIN*SEC	1.000	1.000	0.091	12
4	CG*PERS*NLP*RTF*~AA*MULTI*~DOCX*LANG*~W ORK*INSIGHT*STU*~COLLAB*~ADMIN*~SEC	1.000	1.000	0.091	19
5	CG*PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	14
6	CG*PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*WO RK*INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	16
7	CG*PERS*NLP*RTF*AA*~MULTI*DOCX*LANG*WO RK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	9
8	CG*PERS*NLP*RTF*~AA*~MULTI*~DOCX*LANG*W ORK*INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	18
9	CG*PERS*~NLP*RTF*AA*MULTI*DOCX*LANG*WO RK*INSIGHT*STU*~COLLAB*ADMIN*SEC	1.000	1.000	0.091	1
10	CG*~PERS*NLP*~RTF*AA*MULTI*DOCX*LANG*W ORK*INSIGHT*STU*~COLLAB*~ADMIN*~SEC	1.000	1.000	0.091	3

11 ~CG*~PERS*NLP*RTF*~AA*MULTI*DOCX*~LANG*
WORK*INSIGHT*~STU*COLLAB*ADMIN*SEC

M1 1.000 1.000 1.000

Note- Each configuration of conditions details the presence or absence of each condition where a ~ indicates the absence of a condition and a * indicates the condition in conjunction with what follows in the configuration.

Truth Table Minimization Results for the 10-Condition Intermediate Model

		inclS	PRI	covS	cases
1	CG*~PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*~INSIGHT*STU*~COLLAB*~ADMIN*~SEC	1.000	1.000	0.273	
2	CG*PERS*~NLP*~RTF*AA*~MULTI*DOCX*LANG*~ WORK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.182	
3	CG*PERS*~NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*INSIGHT*STU*COLLAB*~ADMIN*SEC	1.000	1.000	0.091	
4	CG*PERS*NLP*RTF*~AA*MULTI*~DOCX*LANG*~W ORK*INSIGHT*STU*~COLLAB*~ADMIN*~SEC	1.000	1.000	0.091	
5	CG*PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*~W ORK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	
6	CG*PERS*NLP*RTF*~AA*MULTI*DOCX*LANG*WO RK*INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	
7	CG*PERS*NLP*RTF*AA*~MULTI*DOCX*LANG*WO RK*~INSIGHT*STU*~COLLAB*ADMIN*~SEC	1.000	1.000	0.091	
	M1	1.000	1.000	0.909	

Note- Each configuration of conditions details the presence or absence of each condition where a ~ indicates the absence of a condition and a * indicates the condition in conjunction with what follows in the configuration.

Appendix B- QCA Label Definitions

Definitions of QCA Conditions

Code	Variable Name	Definition
CG	Content_Generation	The tool automatically produces instructional materials, resources, or student-facing content.
PERS	Personalization	Tailors content, tasks, or pathways to individual learner profiles or performance data.
NLP	NLP_Conversation	Uses natural language processing for dialogue-based tasks or semantic understanding of user input.
RTF	Realtime_Feedback	Provides instant or dynamic feedback based on user actions or input.
AA	Assessment_Automation	Automates any part of assessment, grading, or feedback at scale.
MULTI	Multimodal_Input	Accepts and processes multiple forms of input (text, images, audio, PDFs, etc.).
DOCX	Document_Transformation	Analyzes or restructures documents; creates new formats or interpretations from existing text.
LANG	Language_Tools	Translates, adjusts reading levels, or supports text-to-speech / speech-to-text.
WORK	Workflow_Integration	Integrates with existing classroom platforms, apps, or digital ecosystems.
INSIGHT	Instructional_Insight	Extracts learning patterns, misconceptions, or analytics useful for instruction.
STU	Student_Creation_Tools	Allows students to create original content using AI (e.g., essays, games, flashcards, videos).
COLLAB	Collaboration_Features	Supports real-time or asynchronous collaboration between users.
ADMIN	Administrative_Support	Automates non-instructional tasks such as communication, planning, and documentation.
SEC	Security_Integrity	Ensures academic integrity, authorship transparency, or data security.